PROBLEM SET 6

Revision Problem set with complex problems

Solutions

1. P = A C + B C + A B D

2. V = X (D1 D2+MD) + Y (D1 + D2) MD + Z D1 D2 MD

3. A = PE LR’ + PP’ LR’ (SF+SP) + PE LR FG + PP’ LR SF FG

4. Y = CL (QE + QG LG)’ + QE’ QG’ BU + BU CM

5. P = D_1 D_0 S_1 + (D_1 + D_0) S_1 S_0

6. C = TL’ M_L + TH, H = T_L M_L’ + TH’ MH

   FL = TL ML’ MH’ + TL’ TH’ MH + TH ML
   FH = TL MH + TH ML’

7. H = TL PH’, V = PL TL’, A = PH TH + PL TL
8. Solution:

(a)

	B_3 B_2					B_1 B_0			
P Q \ R S	0 0	0 1	1 1	1 0		0 0	0 1	1 1	1 0
0 0	11	11	0 1	0 1		11	10	1 0	0 1
0 1	0 1	1 0	10	1 0		1 1	0 0	10	0 1
1 1	0 0	0 0	1 1	1 1		0 1	0 0	0 1	0 0
1 0	1 0	0 1	0 0	0 0		1 1	0 0	1 0	1 1

PQRS

B_3 B_2 \ B_1 B_0	0 0	0 1	1 1	1 0
0 0	1101	1100	1010	1011
0 1	1001	0010	0100	0011
1 1	1110	1111	0000	0001
1 0	0101	0110	1000	0111

(b) B_3 = Q’R’S’ + P’Q’R’ + QR, B_2 = P’R’S’ + PQR + Q’R’S + P’Q’,
B_1 = P’R’S’ + P’Q’R’ + Q’R’S’ + P’RS + PQ’R, B_0 = PQRS + P’S’ + Q’S’ + R’S’.

9. (a) K-map: Z_3 Z_2 Z_1 Z_0

Y_1Y_0 / X_1X_0	0 0	0 1	1 1	1 0		0 0	0 1	1 1	1 0
0 0	0 0	0 0	0 1	0 1		1 0	1 1	0 1	0 0
0 1	0 0	01	1 0	0 1		1 0	0 0	0 0	1 0
1 1	0 0	0 1	1 1	1 0		1 0	1 0	1 0	1 0
1 0	0 0	0 1	1 0	1 0		1 0	0 1	1 1	0 0

(b) $Z_3 = X_1Y_1 + X_0Y_1Y_0$, $Z_2 = X_1X_0Y_0 + X_1'Y_0'Y_1 + X_0Y_1'Y_0 + X_1'Y_1Y_0' + X_1Y_1'Y_0$,

$$Z_1 = X_1'X_0'Y_1' + X_1Y_1Y_0 + Y_1'Y_0' + X_1X_0 + X_0Y_0', \quad Z_0 = X_0'Y_0.$$